KARAN JAGANI

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Profile

Computer Science graduate with hands-on experience in full-stack development using **TypeScript, React, Next.js, Python, FastAPI and PostgreSQL**. Experienced in building end-to-end web applications, REST APIs and database-driven systems, with additional experience in **AI, machine learning and computer vision**. Passionate about becoming a well-rounded software engineer, building strong expertise across the full stack and exploring how **AI and intelligent systems can be integrated into modern software applications**.

Educational Qualification

Anglia Ruskin University
BEng. (HONS) Computer Science, First Class Honours(1:1) Sep 2023 – July 2026

Modules: Software Engineering, Database Design & Implementation, Software Principles, Data Structures & Algorithm Analysis, Artificial Intelligence, Image Processing, Embedded Computing.

Technical Skills

Frontend & Web: TypeScript, React, Redux Toolkit, HTML5, CSS3, REST APIs, Material UI, Data Visualization (Nivo Charts, MUI DataGrid)
Backend & Databases: Node.js, Express.js, MySQL, SQL
Programming Languages: Java, C++, C#, Python
Concept & Core Competencies: OOP, Data Structures & Algorithms, Operating Systems
AI & Machine Learning: Basic knowledge of Machine Learning models and Neural Networks; familiar with Python libraries (NumPy, Pandas, TensorFlow, PyTorch)
Tools & Frameworks: MATLAB, Git/GitHub, Visual Studio, VS Code

Projects

WattFlow – Full stack Energy Broker Management Platform

Next.js • React • TypeScript • Python • FastAPI • PostgreSQL • SQLAlchemy • TailwindCSS • Docker

- Developed a full stack energy broker workflow platform for managing business customers, electricity consumption data, tenders, simulated supplier quotes, contracts and upcoming renewals.
- Built a Next.js/TypeScript frontend and FastAPI/Python REST API, with PostgreSQL persistence through SQLAlchemy and version -controlled database Migration using Alembic.
- Built CSV electricity-consumption import and visualisation, tender and quote-comparison workflows, contract tracking, and a portfolio dashboard highlighting renewals and suggested actions.
- Added automated API workflow testing covering authentication, permissions, organisation isolation, CSV imports, tenders and contract selection, with Docker Compose support for reproducible local development.

E-commerce Admin Panel (SERN Stack) – Admin Dashboard (Full-stack Project)

SQL (MySQL) • Express • React • Node.js | Vite • Redux Toolkit • Material UI (MUI) • Nivo Charts
Aug. 2023 – Oct. 2023

- Built a responsive E-commerce admin panel using the **SERN** stack (MySQL, Express, React, Node.js) to manage products, user and admin operation e
- nd-to-end.

- Developed and integrated Rest **APIs** + **MySQL** database for secure, reliable **CRUD** workflows (inventory, user accounts, and data management).
- Implemented a modern UI with **React** (Vite), **Redux Toolkit**, Material UI/Emotion, including reusable components and optimized **mobile/desktop** layouts.
- Delivered analytics & reporting using **Nivo Charts** and **MUI** Data Grid with **Sorting**, **Filtering**, and **pagination** for fast insights and scalable tables.

AI-Based Skin Image Analysis for Neonatal Jaundice Detection

Independent Research Project – 2026

- Developing a vision-based system to estimate skin colour for neonatal jaundice detection using **MATLAB** and **machine learning**.
- Built an image-processing pipeline to remove pores, hairs, and blemishes before colour classification.
- Prototyped algorithms in MATLAB and **deployed the system on a Raspberry Pi** using **OpenCV** for real-time image capture and analysis.

Multimodal AI & Explainable Machine Learning (Tabular-Image-Audio-NLP)

Artificial Intelligence Module Project Sep. 2024 – Nov. 2024 ARU, Cambridge

- Developed **end-to-end** AI pipeline across four **data modalities** using **Python** & Jupyter notebook.
- Built and compared **classical models** (e.g. Logistic regression, SVM, Naive Bayes, Random Forest) **DL models** (ANN/CNN) and a **fine-tuned Transformer** (e.g., Distilbert) for text classification.
- Applied **Explainability** & **interpretability** methods (**SHAP**, **Grad-CAM**) to analyse feature/word importance and visual model decision-making for improved transparency.
- Performed **unsupervised learning model** (e.g., **K-means / LDA topic modelling**) and compared discovered patterns with supervised results; evaluated using **Precision, recall, F1, ROC-AUC** with clear visualizations (confusion matrix, ROC curves).
- Completed full project cycle: requirement analysis, implementation, UI design, testing, and maintenance planning, following structured software engineering principles.

Work Experience

Customer Assistant (Sainsbury's, Cambridge, UK)
Oct 2023 – Present

- Operate tills efficiently, assisting customers with purchases and providing excellent service.
- Work both independently and within a team, using technology and initiative to deliver a smooth shopping experience.

Student Ambassador (Anglia Ruskin University)
Oct 2023 – March 2026che

- Represent ARU at Applicant Days and Open Days, supporting event logistics, visitor engagement, and campus tours.
- Provide information about university life, accommodation, and academic programs, ensuring a welcoming and professional experience.

Visa Administrator (MyCalculus Visa Consultant services, Rajkot, India)
Jan 2023 – July 2023

- **Processed** visa **applications**, **ensuring** compliance with immigration **regulations** and accurate documentation.
- Verified client records and maintained data accuracy across internal systems.

Languages

- **English**: Fluent, **Gujarati**: Native, **Hindi**: Conversational